FastFill: Experiments and Evaluation

实验与评测（中英双语简稿）

20 October 2026

Experiments and Evaluation

Task and protocol. Given room geometry, fixed objects and a furniture-category list, FastFill estimates furniture dimensions from a training-only prior and uses a 27B placement model to predict positions, orientations and support relations. Dimensions remain fixed during deterministic repair (S1) and optional local-LLM refinement (O2). We report the model, repair and refinement stages separately on the same 8,647 test requests. These are descriptive results on a previously exposed test split, not a new blind benchmark. The original matched 8B joint-size/placement study and its registered endpoints remain a separate experiment.

Metric selection. Prior work combines distributional image quality with task-specific checks: ATISS and DiffuScene report rendering-based distributional metrics; LayoutGPT also reports room-boundary violations; OptiScene combines FID, object overlap and visual/human judgments [1–4]. Accordingly, we emphasize *geometric success with a size-error check*, and report downstream admission and visual quality separately. Table 1 fixes the terminology and denominators; no metric is retrospectively promoted to a preregistered endpoint.

Table 1: Metrics and reporting rules. N denotes all requested rooms.

Metric	Definition and denominator
Geometry success (GSR) ↑	Complete output passing the frozen checker: boundary, support, ceiling, main/severe collision and fixed-object blocking checks. Rate over N; failures count as zero.
Size log-MAE ↓	Mean absolute log dimension error against reference dimensions, averaged over three axes, objects, then complete rooms. This checks size estimation, not placement.
Complete rate ↑	All requested objects returned with schema-valid geometric fields; denominator N.
Object overlap (OOR) ↓	Pairwise overlap area divided by total object footprint area; mean over complete rooms, with completeness reported alongside.
Strict admission ↑	Complete output satisfying the local downstream endpoint, attachment and admission predicates; denominator N. This is an application-specific automatic metric.
Visual / mesh quality	FID ↓, KID ↓ on a shared rendered subset; voxel mesh collision-free rate ↑ on successfully analyzed rooms. Always report coverage.

For a complete room r with n_r objects, the two continuous geometry metrics are

$$E_r = \frac{1}{3n_r} \sum_{i=1}^{n_r} \sum_{d\in\{x,y,z\}} \left| \log \frac{\hat{s}_{rid}}{s_{rid}} \right|, \qquad \mathrm{OOR}_r = \frac{\sum_{i<j} |A_{ri} \cap A_{rj}|}{\sum_i |A_{ri}|}. \tag{1}$$

Here A_{ri} is the axis-aligned footprint formed from the predicted center and width/depth *without applying yaw*. All movable pairs enter OOR, without support or height exemptions. OOR is neither an out-of-room rate nor

a collision probability, and can exceed one. GSR instead uses oriented geometry with fixed tolerances and semantic collision exemptions. Strict admission uses its own fixed adaptation, including boundary snapping even for the row labeled "raw"; it is neither a RoomGenBench standard nor the human usability rate (UR) in OptiScene. Human UR has not been measured.

Full-test results. Table 2 compares two refinement triggers. The old rule refines checker failures; the new rule refines failures of either the checker or strict admission. Both reuse the same 27B predictions and S1 outputs.

Table 2: Stage comparison on the same 8,647 requests. Rates are percentages; OOR and size error are means over complete outputs.

Stage	Complete ↑	GSR ↑	Strict ↑	OOR ↓	Size error ↓
27B-P raw	99.95	44.34	21.61	0.2560	0.2824
27B-P + S1	99.95	74.22	29.13	0.1526	0.2824
+ O2: checker trigger	100.00	89.01	34.35	0.1081	0.2824
+ O2: checker/strict trigger	100.00	89.55	49.42	0.0996	0.2824

The new trigger increases strict admission by 15.07 percentage points (95% CI [14.20, 15.95]) over the old trigger; GSR increases by 0.53 points (95% CI [0.09, 0.98]). Intervals use paired resampling of 3,056 house components, 2,000 draws and seed 42, without multiplicity correction; they do not measure training-seed variability. The new system refines 6,155/8,647 rooms (71.18%), using 11,578 local O2 generations and zero external API calls. Recorded O2 processing totals approximately 46.0 GPU-hours, including model loading and a recovered shard, but excluding the reused placer inference. Size log-MAE is 0.2824 for both final systems: the trigger change improves placement without changing requested dimensions.

Separate visual evaluation. Table 3 concerns the *old checker-trigger system*, not the new trigger. Reference and generated scenes share assets, cameras and rendering settings. FID/KID use 2,048-dimensional Inception features and one fixed perspective per room. Of 1,009 selected rooms, 1,005 are successfully rendered/analyzed; four lack a required asset. Mesh collision-free means no eligible non-parent movable-object pair exceeds 500 cm^3 voxel overlap (2 cm voxel pitch), rather than exact zero intersection.

Table 3: Historical visual subset (1,005 rooms). S uses the old checker trigger. KID is on its raw scale; ± is its recorded standard error. Mesh brackets are 95% CIs.

Method	FID ↓	KID ↓	Mesh CF@500 (%) ↑
27B-P raw (A)	22.883	0.000678 ± 0.000140	57.51 [53.36, 61.32]
Checker-trigger system (S)	22.786	0.000700 ± 0.000142	77.51 [74.30, 80.62]

The small FID difference and slightly higher KID do not establish a distributional-quality improvement. Visual scores are conditional on this subset and cannot be assigned to the new full-test system. Published paper numbers are not direct baselines because datasets, conditioning and rendering differ. These tables support stage-wise system evaluation, not a claim of superiority to prior methods; independent blind testing and human usability assessment remain outstanding.

实验与评测

任务与评测设置。 输入为房间几何、固定物体和家具类别清单。FastFill 先使用仅从训练集拟合的先验估计尺寸，再由 27B 放置模型预测位置、朝向和支撑关系。确定性修复 S1 和可选的本地大模型 O2 优化均保持请求尺寸不变。我们在相同的 8,647 个测试请求上分别报告模型、修复及优化阶段。结果来自已曝光的历史测试集，属于描述性评估；原始 8B 联合尺寸预测与分阶段放置对照及其预注册终点仍是独立实验。

指标选择。 相关论文通常同时报告分布质量和任务指标：ATISS、DiffuScene 使用渲染分布指标，LayoutGPT 还报告房间越界率，OptiScene 结合 FID、物体重叠和视觉 / 人工评价 [1–4]。本节重点展示**几何成功率与尺寸误差**，将下游准入和视觉质量分别报告。表 4 固定指标名称、分母与失败处理；这种写作安排不改变原先的预注册终点。

表 4: 指标定义及统计口径。N 为全部请求房间数。

指标	定义及分母
几何成功率 GSR ↑	输出完整，且通过冻结检查器的边界、支撑、层高、主要 / 严重碰撞及固定物阻挡检查；全部 N 个请求为分母，失败计零。
尺寸 log-MAE ↓	相对标注尺寸的绝对对数误差，依次对三轴、物体及完整房间取均值；仅衡量尺寸估计。
完整输出率 ↑	返回全部请求物体且几何字段满足格式要求；分母为 N。
物体重叠比 OOR ↓	所有物体对的交叠面积之和除以总物体占地面积；对完整房间取均值，必须同时报告完整率。
严格准入率 ↑	完整输出同时满足本地下游 endpoint、attachment 和 admission 判定；全部请求为分母，属于应用侧自动指标。
视觉 / 网格质量	在共同渲染子集上报告 FID ↓、KID ↓；在成功分析的房间上报告网格无超阈值碰撞率 ↑，并列出覆盖率。

对含 n_r 个物体的完整房间 r，尺寸误差和重叠比为

$$E_r = \frac{1}{3n_r}\sum_{i=1}^{n_r}\sum_{d\in\{x,y,z\}}\left|\log\frac{\hat{s}_{rid}}{s_{rid}}\right|, \qquad \mathrm{OOR}_r = \frac{\sum_{i<j}|A_{ri}\cap A_{rj}|}{\sum_i |A_{ri}|}. \tag{2}$$

其中 A_{ri} 是由预测中心和宽深构造、**忽略 yaw** 的轴对齐矩形。OOR 纳入全部可移动物体对，不豁免高度分离或支撑关系；它不是越界率，也不是碰撞概率，数值可能大于 1。GSR 则采用旋转几何、固定容差和语义碰撞豁免。严格准入使用独立的固定适配步骤，包括在 “raw” 行也执行的边界吸附；它既不是 RoomGenBench 的标准指标，也不等于 OptiScene 的人工可用率 UR。本项目尚未测量人工 UR。

全量测试结果。 表 5 对比两种优化触发规则：旧规则仅对检查器失败的房间调用 O2；新规则对检查器或严格准入任一失败的房间调用 O2。两者复用相同的 27B 原始预测和 S1 输出。

表 5: 相同 8,647 请求的阶段对照。比例单位为百分比；OOR 与尺寸误差仅对完整输出取均值。

阶段	完整率 ↑	GSR ↑	严格准入 ↑	OOR ↓	尺寸误差 ↓
27B-P 原始输出	99.95	44.34	21.61	0.2560	0.2824
27B-P + S1 修复	99.95	74.22	29.13	0.1526	0.2824
+ O2：检查器触发	100.00	89.01	34.35	0.1081	0.2824
+ O2：检查器 / 严格准入触发	100.00	89.55	49.42	0.0996	0.2824

新规则相对旧规则使严格准入率提高 15.07 个百分点，95% 区间为 [14.20, 15.95]；GSR 提高 0.53 个百分点，区间为 [0.09, 0.98]。区间按 3,056 个房屋连通组配对 bootstrap，重采样 2,000 次、seed=42，未经多重比较校正，也不包含训练随机种子的变异。新系统对 6,155/8,647 间房间（71.18%）执行优化，共 11,578 次本地 O2 生成，

外部 API 调用为零。O2 实测计算量约为 46.0 GPU 小时，包含模型加载和恢复分片，不包含复用的放置模型推理。两种最终系统的尺寸 log-MAE 均为 0.2824；触发规则改变没有修改请求尺寸。

独立视觉子集。 表 6 对应**旧检查器触发系统**，不能归入新规则的全量成绩。参考场景与生成场景使用相同资产、相机和渲染设置；FID/KID 使用 2,048 维 Inception 特征，每房间一个固定视角。选定的 1,009 间房间中，1,005 间完成渲染和分析，4 间因所需资产缺失而未纳入。网格无超阈值碰撞指任一非直接父子关系的可移动物体对，其体素交叠体积不超过 500 cm^3，体素边长为 2 cm；它不表示精确零相交。

表 6: 历史视觉子集（1,005 房间）。S 使用旧检查器触发规则。KID 不缩放，± 为记录的标准误；网格指标方括号为 95% 区间。

方法	FID ↓	KID ↓	网格 CF@500 (%) ↑
27B-P 原始输出 (A)	22.883	0.000678 ± 0.000140	57.51 [53.36, 61.32]
旧触发系统 (S)	22.786	0.000700 ± 0.000142	77.51 [74.30, 80.62]

FID 的微小差异与略高的 KID 不足以证明分布质量改善。这些视觉结果仅适用于该子集。论文已发表数字因测试数据、输入条件和渲染设置不同，不能直接作为本表对照。当前证据支持系统各阶段的效果分析；对既有方法的优越性、独立盲测表现及人工可用性仍需额外验证。

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